Problem 1. During one day Sarah receives X legitimate emails and Y spam emails. Suppose X and Y are independent, $X \sim Pois(\lambda_x)$ and $Y \sim Pois(\lambda_y)$. Assuming one day Sarah receives n emails. Given this information what is the expected value of how many legitimate email she has? Show all the work.

Problem 2. Use the Central limit theorem to approximate the following probabilities.

(a) Suppose we roll a fair 7-sided die until we get 50 seven's. What is the probability it takes at least 500 rolls until this happens? (use continuity correction for this part.)

(b) Let X be the sum of 300 real numbers, and Y be the same sum, but with each number rounded to the nearest integer before summing. If the rounded offs (errors) are independent and each one is uniformly distributed over (-0.1, 0.1), use the Central Limit Theorem to estimate the probability that $P(|X - Y| > 30)$.

Problem 3. Let $X_1, \cdots, X_n$ be a random sample from

$$f_\theta(x) = \begin{cases} (\theta - 1)^2 \dfrac{ln(x)}{x^\theta} & x > 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) In this part, assume $\theta > 1$. Find the maximum likelihood estimator (MLE) for θ.

(b) In this part, assume $\theta > 2$. Find the method of moments estimator (MOM) for θ.

Problem 4. Let $X_1, \cdots, X_n$ be a be a random sample from

$$f_\theta(x) = \begin{cases} \dfrac{1}{\theta}\, x^{\dfrac{1-\theta}{\theta}} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the maximum likelihood estimator (MLE) for θ.

(b) Find the method of moments estimator (MOM) for θ.

Problem 5. Let X and Y be independent continuous random variables each uniformly distributed on the interval [0,1], and let $Z = max\{2X, Y\}$. Find the pdf of $f_Z(z)$.

Problem 6. Let $X \sim N(0,1)$. Let Y= g(X) where

$$g(t) = \begin{cases} -t & \text{if } t \leq 0 \\ \sqrt{t}, & \text{if } t > 0 \end{cases}$$

Find the probability density function of Y.

Problem 7. A fair die is tossed twice, independently. Let the random variable Z be defined as the number of dots seen on the first toss minus the number of dots seen on the second toss. Find the expected value of Z^2.

Problem 8. Let $X_1, X_2, \cdots, X_n$ be independent identically distributed random variables where each $X_i \sim Exp(\lambda_i)$. Let $L = min(X_1, X_2, \cdots, X_n)$. Find pdf of L. Show all your work.

Problem 9. A group of 20 coders from 10 different tech startups participates in a hackathon (2 from each startup). During the team-building session, the participants are randomly paired into 10 breakout rooms, each accommodating 2 coders. Let X be the number of breakout rooms where both coders are from the same startup. Find the Var(X).

Problem 10. People are arriving at a party one at a time. While waiting for more people to arrive they entertain themselves by comparing their birthdays. Let X be the number of people needed to obtain a birthday match, i.e., before person X arrives there are no two people with the same birthday, but when person X arrives there is a match. For example, $X = 5$ would mean that the first four people to arrive all have different birthdays, but the fifth person to arrive matches one of the first four. Find $P(X = 3 \text{ or } X = 4)$.

Problem 11. Consider 10 independent tosses of a biased coin with the probability of Heads at each toss equal to p , where $0 < p < 1$.

(a) Let A be the event that there are 6 Heads in the first 8 tosses. Let B be the event that the 9th toss results in Heads. Find $P(B|A)$.

(b) Find the probability that there are 3 Heads in the first 4 tosses and 2 Heads in the last 3 tosses.

(c) Given that there were 4 Heads in the first 7 tosses, find the probability that the 2nd Heads occurred at the 4th toss.

(d) Calculate the probability that there are 5 Heads in the first 6 tosses and 3 Heads in the last 5 tosses.

Problem 12. Each of people puts his or her name on a slip of paper (no two have the same name). The slips of paper are shuffled in a hat, and then each person draws one (uniformly at random at each stage, without replacement). Find the mean and standard deviation of the number of people who draw their own names.